

✓✓Documents

Document	Text
D1	students watch youtube
D2	youtube watch youtube
D3	students write comments
D4	youtube write comments

→ (3, 5)
→ (3, 5)

Step 1: Create the Corpus

Combine all documents:

students watch youtube
youtube watch youtube
students write comments
youtube write comments

Step 2: Build the Vocabulary

Find all unique words:

students
watch
youtube
write
comments

So,

Vocabulary Size (V) = 5

students	watch	youtube	write	comments
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Step 3: One-Hot Encode Each Word

Word	One-Hot Vector
students	[1, 0, 0, 0, 0]
watch	[0, 1, 0, 0, 0]
youtube	[0, 0, 1, 0, 0]
write	[0, 0, 0, 1, 0]
comments	[0, 0, 0, 0, 1]

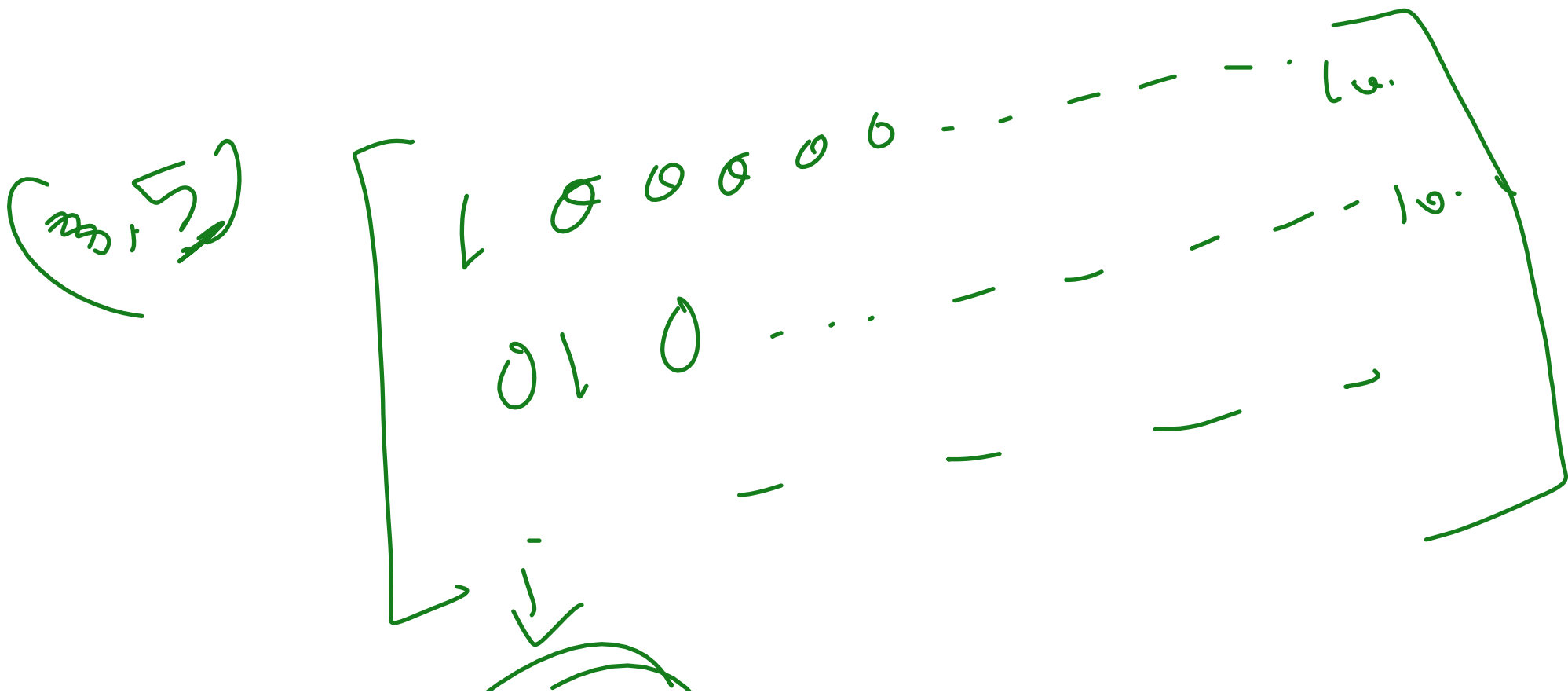
Example Encoding of D1

D1 =

students watch youtube

Becomes

students → [1, 0, 0, 0, 0]
watch → [0, 1, 0, 0, 0]
youtube → [0, 0, 1, 0, 0]



Con

- 1) sparsity ^(toos) y slow computation
- 2) size mismatch
- 3) out of vocabulary problem
- 4) semantic meaning ~~X~~

Cat $\rightarrow [1 \ 0 \ 0]$
dog $\rightarrow [0 \ 1 \ 0]$
con $\rightarrow [0 \ 0 \ 1]$

King
Queen

